The IoT Academy
Connecting The Unconnected

EMBEDDED SYSTEMS & IoT

Certification Course

READY TO TRANSFORM YOUR CAREER?

4 Months



In Collaboration With

futureskills prime
A MeitY - nasscom Digital Skilling Initiative

nasscom

ISO 27001 CERTIFIED

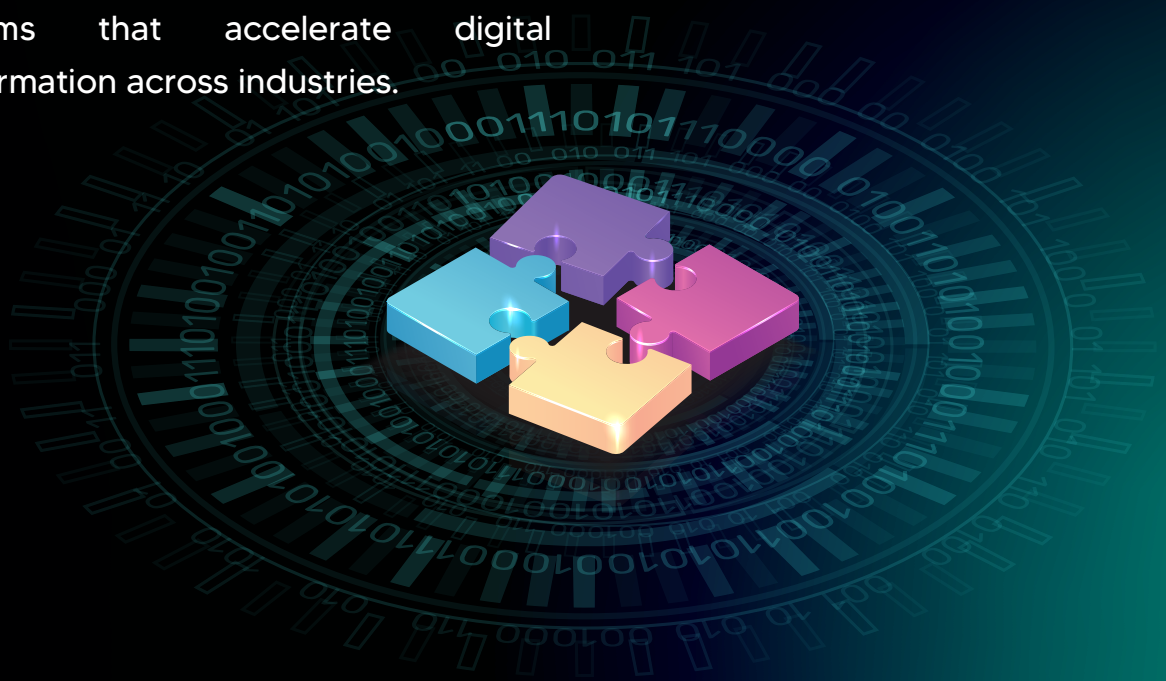
About

ABOUT NASSCOM AND FUTURESKILLS

NASSCOM (National Association of Software and Service Companies)

NASSCOM is India's leading industry association for the IT, IT-enabled services, and emerging technology sectors, committed to driving innovation, digital excellence, and global competitiveness. It plays a pivotal role in shaping industry standards, influencing policy, and fostering collaboration between government, academia, and enterprises. Through its initiatives, NASSCOM empowers India's tech ecosystem with future-ready talent, research insights, and programs that accelerate digital transformation across industries.

FutureSkills Prime is a digital skilling initiative by NASSCOM, developed in partnership with the Government of India, aimed at building a future-ready workforce. It offers industry-vetted courses, certifications, and learning pathways in emerging technologies like AI, Data Science, Cloud, and Cybersecurity to help professionals upskill and stay competitive.



About

THE IOT ACADEMY

The IoT Academy, established in **2017** is a fast emerging company imparting quality programs for skills training, internship and guidance in cutting edge technologies like **Data Science, Machine Learning, Artificial Intelligence, Internet of Things, Embedded Systems & many more** and focused on helping people develop the skills they need to thrive in the rapidly growing digital economy.

The IoT Academy has collaborated with various premier institutes e.g. **E&ICT Academy, IIT-Guwahati, IIT-Roorkee and IIT-Kanpur** for **Advance Certification courses** to take provide outcome-centric solutions to help them achieve their professional goals.

400+
Hiring Partners

45000+
Learners

8+ Years
Of excellence

81%
Salary Hike

91%
Happy Outcomes

Quality education and career opportunities shouldn't come at a high cost. While many ed-tech companies prioritize profits, **The IoT Academy** is committed to transforming students' lives. We make premium professional education accessible to the masses at just 10% of the market cost.



The IoT Academy
Connecting The Unconnected

Forbes INDIA

We are proud to share that

The IoT Academy

has been featured in **Forbes India** under

- ✦ "Visionary Companies Building India's Future"
- ✦ Ranked #2 Nationwide

This recognition reflects our unwavering mission to empower learners with future-ready skills!

- ✦ From classrooms to careers...
- ✦ From learners to innovators...
- ✦ From ambition to achievement...

A heartfelt thank you to our students, mentors, industry partners, and community for trusting us and growing with us.

The journey has just begun.
Bigger milestones ahead.



AWARDS & RECOGNITIONS













AWARDS & RECOGNITIONS













FACULTY DEVELOPMENT PROGRAMS (FDPS)



- Conducted multiple intensive workshops for college faculty on AI-driven signal processing techniques.
- Delivered hands-on training using Edge Impulse, Python, and Hugging Face to build and deploy real-world AI models.
- Empowered educators to seamlessly integrate advanced AI/ML algorithms into their teaching and research.

INDUSTRIAL VISITS FOR STUDENTS

- Organized immersive visits to leading IoT manufacturing facilities and smart-factory labs.
- Enabled learners to observe real-world deployment of embedded systems, edge computing, and sensor networks.
- Fostered practical insights through direct interaction with industry engineers and live project demonstrations.



WHY THE IoT ACADEMY IS THE RIGHT CHOICE FOR YOU?

Climb the Mountain of Success with The IoT Academy

Just like climbing a mountain requires the right guide, equipment, and path - your Embedded & IoT career needs the right mentor, tools, and learning approach. The IoT Academy provides everything you need vfor this transformational journey.



THE IoT ACADEMY
MILESTONES

78% Average Salary Hike	**₹40 LPA** Highest Package	**7000+** Success Stories
95% Placement Rate	**400+** Hiring Partners	**₹21 LPA** Average CTC



Our

ADVANCED EMBEDDED SYSTEMS & IOT LAB








Our state-of-the-art Embedded & IoT Lab is designed to give learners hands-on exposure to the technologies that are shaping the future. The lab is equipped with industry-grade hardware, advanced development boards, real-time sensors, and smart IoT devices, allowing students to build, test, and deploy innovative solutions just like professionals in the field.

The facility provides an end-to-end learning environment—from embedded programming and circuit design to cloud-connected IoT deployments and AI-enabled edge applications.

Why Learn
EMBEDDED SYSTEMS & IoT

- **High Demand:** Over 30 million connected IoT devices by 2026.
- **Wide Opportunities:** Manufacturing, automotive, healthcare, smart homes & agriculture.
- **Growing Importance:** Every industry needs smart, connected products.
- **Problem Solving + Creativity:** Design, build, and deploy real-world systems.
- **Edge Advantage:** Develop low-latency, reliable solutions on resource-constrained hardware.



GLOBAL MARKET OPPORTUNITY

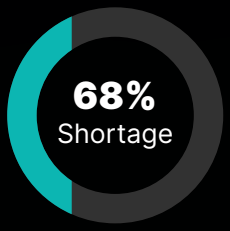
Job Title Recognition:

"Backbone Job of the 21st Century" – Embedded Systems & IoT Engineering

Report By glassdoor

Talent Shortage:

68% shortage of Embedded & IoT professionals.



Market Growth:

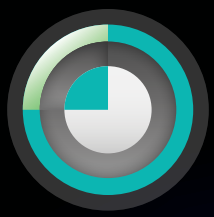
- **33% CAGR** Embedded Systems market in India
- **25–30% annual job growth** through 2030
- Growing **much faster** than all engineering roles

33% Market Growth

25-30% Annual Job Growth

Market Opportunity:

Global Embedded & IoT market to reach **$180B+ by 2032**



$180B+ Market

Supply vs Demand Gap:

Only 7% engineers have advanced Embedded skills
50–70% higher salaries for Embedded, IoT, EV & Robotics roles
80% enterprises to adopt real-time embedded & IoT automation by 2027

Industry Trust:

400+ Hiring Partners | 6200+ Success Stories | 95% Placement Rate

Momentum:

Embedded Systems & IoT market rising at 33% CAGR
Job demand is growing 25–30% annually through 2030

HOW WE MAKE YOU JOB READY



Enroll & Learn from Experts

Work on Live Projects

Build Resume & Digital Portfolio

Get 1:1 Career Assistance

Crack Interviews & Get Placed

THE IoT ACADEMY EDGE

01 400+ Hiring Partners

02 Structured, Industry-Vetted Curriculum

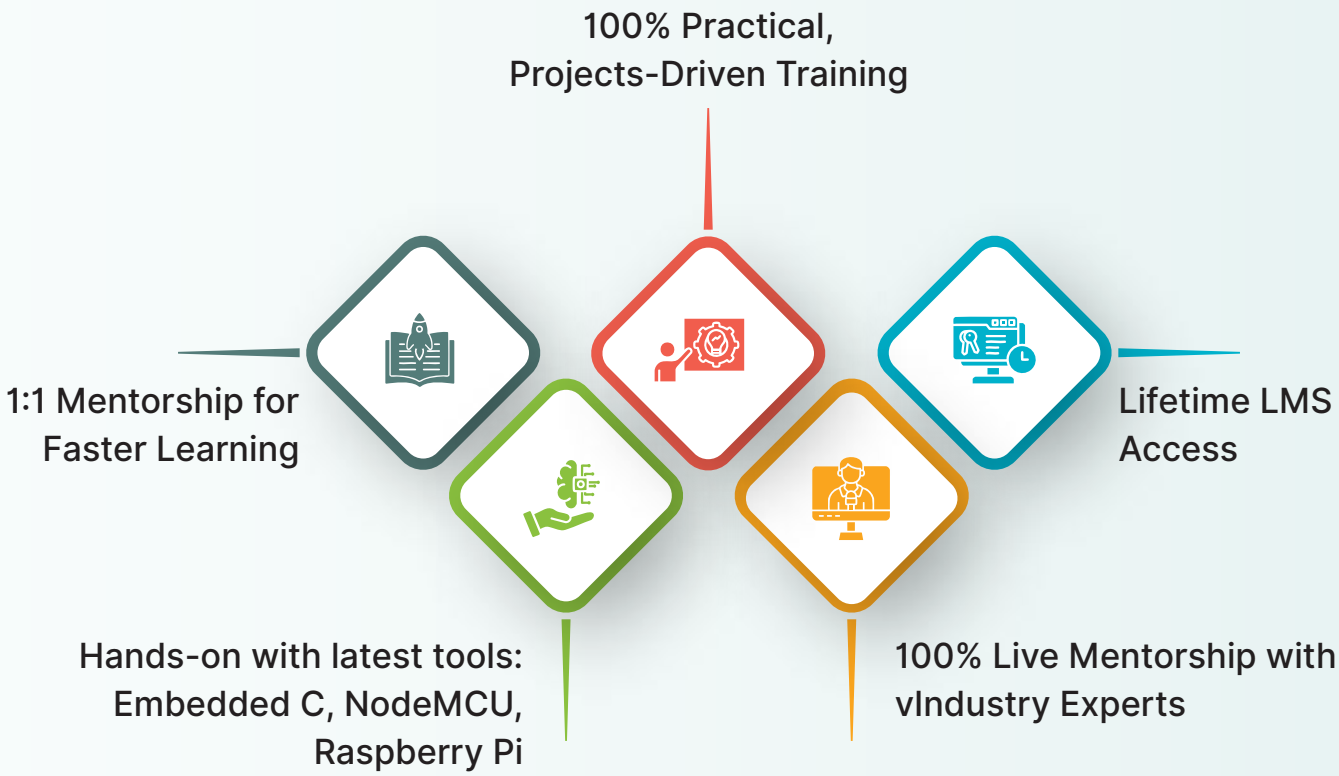
03 Hands-on Practice with Real Campaigns

04 Resume & LinkedIn Optimization

05 12k+ Learners & Alumni Community

06 International Certification

PROGRAM HIGHLIGHTS



PLACEMENTS & HIRING



PROGRAM CURRICULUM

Embedded Systems & IoT





Program Curriculum

MODULE 1

Introduction to Embedded Systems

- Evolution of embedded systems
- Microcontrollers vs Microprocessors
- Embedded applications in automotive, healthcare, consumer electronics, and IoT
- Development tools: IDEs, toolchains, debuggers

Lab: Setting up Arduino/ESP32 or STM32 development board, "Hello World" LED blink.

MODULE 2

Embedded C & Low-Level Programming

- C basics for embedded (data types, pointers, bit manipulation)
- Memory organization (stack, heap, registers)
- I/O mapping and register-level programming

Lab: GPIO programming, reading switches, driving LEDs.

MODULE 3

Interfacing & Peripherals

- ADC/DAC concepts
- Timers, PWM, interrupts
- Serial protocols: UART, I2C, SPI

Lab: Interface a temperature sensor + display output on serial monitor.

Program Curriculum

MODULE 4

Embedded Hardware Basics

- Sensor & actuator overview
- Memory (EEPROM, Flash, SRAM)
- Power supply, clock system, watchdog timers
- Intro to Embedded Linux

Lab: Work with external EEPROM, log sensor data.

MODULE 5

Real-Time Operating Systems (RTOS)

- Multitasking concepts
- Scheduling (preemptive, cooperative)
- Tasks, semaphores, queues, mutexes
- FreeRTOS / Zephyr basics

Lab: Create multiple tasks with priorities on FreeRTOS.

MODULE 6

Embedded Linux

- Bootloaders & kernel basics
- Device drivers introduction
- Cross-compilation, GPIO access from Linux

Lab: GPIO programming on Raspberry Pi.

Program Curriculum

MODULE 7

IoT Communication Protocols

- WiFi, Bluetooth/BLE 5, Zigbee
- LPWAN: LoRaWAN, NB-IoT
- Application protocols: MQTT, CoAP, HTTP

Lab: Send sensor data to MQTT broker using ESP32/STM32 + WiFi.

MODULE 8

IoT Security & OTA Updates

- Security threats in IoT
- Encryption (TLS/DTLS)
- Secure boot & firmware signing
- OTA firmware updates

Lab: Implement a secure MQTT connection with TLS.

MODULE 9

IoT Architecture

- Edge, Gateway, Cloud
- Device identity & provisioning
- Scalability in IoT systems

Lab: Set up an IoT gateway to forward sensor data to cloud.

Program Curriculum

MODULE 10

Cloud Platforms

- AWS IoT Core, Azure IoT Hub, Google IoT
- Open-source platforms (ThingsBoard, Node-RED, Mosquitto)
- REST APIs, WebSockets

Lab: Push IoT sensor data to ThingsBoard dashboard.

MODULE 11

Data Analytics & IoT Dashboards

- Data storage (SQL/NoSQL, Time-series DBs)
- Visualization tools (Grafana, Power BI)
- IoT analytics use-cases

Lab: Build a real-time dashboard for temperature & humidity monitoring.

MODULE 12

Edge AI & TinyML

- Introduction to TinyML (TensorFlow Lite Micro, Edge Impulse)
- Running ML on constrained devices
- Use cases: anomaly detection, predictive maintenance

Lab: Deploy a simple ML model on MCU for gesture recognition.

Program Curriculum

MODULE 13

Low-Power & Energy Harvesting IoT

- Power management techniques
- Battery optimization & estimation
- Energy harvesting (solar, vibration, RF)

Lab: Implement sleep/wake cycles to extend battery life.

MODULE 14

Industrial IoT & Smart Systems

- Industry 4.0 concepts (IIoT)
- PLC/SCADA integration
- Protocols: OPC-UA, Modbus
- Smart home & Matter protocol (latest standard)

Lab: Build a simple smart home setup (e.g., light automation with mobile app).

MODULE 15

Capstone Project Development

- Proposal & system design
- Hardware selection & firmware development
- Cloud integration & security layer

MODULE 16

Final Project Demo & Evaluation

- Testing & debugging
- Performance analysis (latency, power, security)
- Final presentation & submission

ADVANCED TOOLS YOU WILL LEARN

Arduino IDE:
For programming and testing Arduino-based embedded projects.

ESP32/ESP8266 Toolchai:
For developing Wi-Fi-enabled IoT applications.

Raspberry Pi:
For building IoT gateways and running Linux-based applications.

Embedded C/C++ Compilers:
For writing and compiling firmware for microcontrollers.

Keil uVision / MPLAB:
For ARM and PIC microcontroller development environments.

Proteus / Tinkercad:
For circuit simulation and hardware prototyping.

Embedded Linux Tools:
For configuring and programming Linux-based IoT devices.

NodeMCU Platform:
For quick IoT prototyping using Lua or Arduino framework.

MQTT/HTTP Tools:
For testing IoT communication protocols.

Firebase / AWS IoT Core / Azure IoT Hub:
For cloud connectivity and device data management.

Node-RED:
For building IoT workflows and dashboards without heavy coding.

Sensors & Actuator Kits:
For integrating temperature, motion, humidity, and electrical sensors.

Serial Monitor & Debugging Tools:
For testing firmware and device communication.

SKILLS YOU WILL MASTER

Embedded Programming:
Writing firmware in C/C++ for microcontrollers.

Circuit Design:
Creating and testing electronic circuits for embedded systems.

Sensor Integration:
Connecting and calibrating sensors for real-time data acquisition.

Microcontroller Interfacing:
Working with GPIO, ADC, PWM, UART, SPI, and I2C.

IoT Architecture Understanding:
Learning device, gateway, and cloud communication layers.

Wireless Communication:
Implementing Wi-Fi, Bluetooth, and RF modules in IoT systems.

IoT Protocols:
Using MQTT, HTTP, and TCP/IP for device communication.

Cloud Connectivity:
Sending and storing IoT data on cloud services.

Real-Time Data Monitoring:
Building dashboards to track live sensor values.

Embedded Linux Basics:
Working with Linux for IoT and automation tasks.

Troubleshooting & Debugging:
Identifying and fixing hardware, firmware, and communication issues.

PCB Basics:
Understanding printed circuit board design and layout fundamentals.

Project Development:
Building end-to-end embedded and IoT solutions for real-world applications.

Live Industry-Based Projects

Industry projects are an important part of the Embedded Systems & IoT. These projects let you work on real problems from real companies. You get hands-on experience in Embedded Systems. This helps you learn practical skills and solve tough problems in different industries.

Smart Home Automation System

Description: Controls home appliances remotely using sensors and IoT connectivity.

Tools: ESP8266/ESP32, Relays, MQTT, Mobile App.

Outcomes: Students build a complete IoT-based home automation prototype.



IoT-Based Smart Irrigation System

Description: Soil moisture data drives automatic water pump activation.

Tools: Arduino/ESP32, Soil Moisture Sensor, Relays, Blynk App.

Outcomes: Students build a water-saving smart agriculture module.



Live Industry-Based Projects

Home Security Alarm System

Description: Detects intrusion using motion sensors and triggers alarms/notifications.

Tools: PIR Sensor, Arduino, GSM Module.

Outcomes: Fully functional home security alert device.



Automatic Street Light Control

Description: Lights turn ON/OFF based on ambient light using LDR sensors.

Tools: Arduino, LDR, Relays.

Outcomes: Energy-efficient automated lighting system.



Smart Parking System

Description: Detects empty parking slots and updates availability to cloud.

Tools: Ultrasonic Sensors, ESP8266, Web App.

Outcomes: A real-time parking availability system.



Live Industry-Based Projects

Fire & Gas Detection System

Description: Identifies fire or gas leaks and sends emergency alerts.

Tools: MQ-2 Gas Sensor, Flame Sensor, Arduino, GSM Module.

Outcomes: A safety alert device for homes and industries.



Smart Water Level Controller

Description: Automatically controls water pump based on tank water level.

Tools: Ultrasonic Sensor, Arduino, Relays.

Outcomes: Hands-on automation system reducing manual pump monitoring.



IoT-Based Asset Tracking

Description: Tracks location of packages or vehicles using GPS + cloud.

Tools: GPS Module, ESP32, Cloud IoT Dashboard.

Outcomes: live GPS-tracking prototype.



Live Industry-Based Projects

Weather Monitoring Station

Description: Collects temperature, humidity, and pressure and displays/upload data to cloud.

Tools: DHT22/BMP280, ESP32, ThingSpeak.

Outcomes: A working weather station with real-time cloud dashboard.



Digital Door Lock System

Description: Unlocks door using password/PIN via keypad module.

Tools: Arduino, Keypad, Servo Motor.

Outcomes: A secure electronic door lock system.

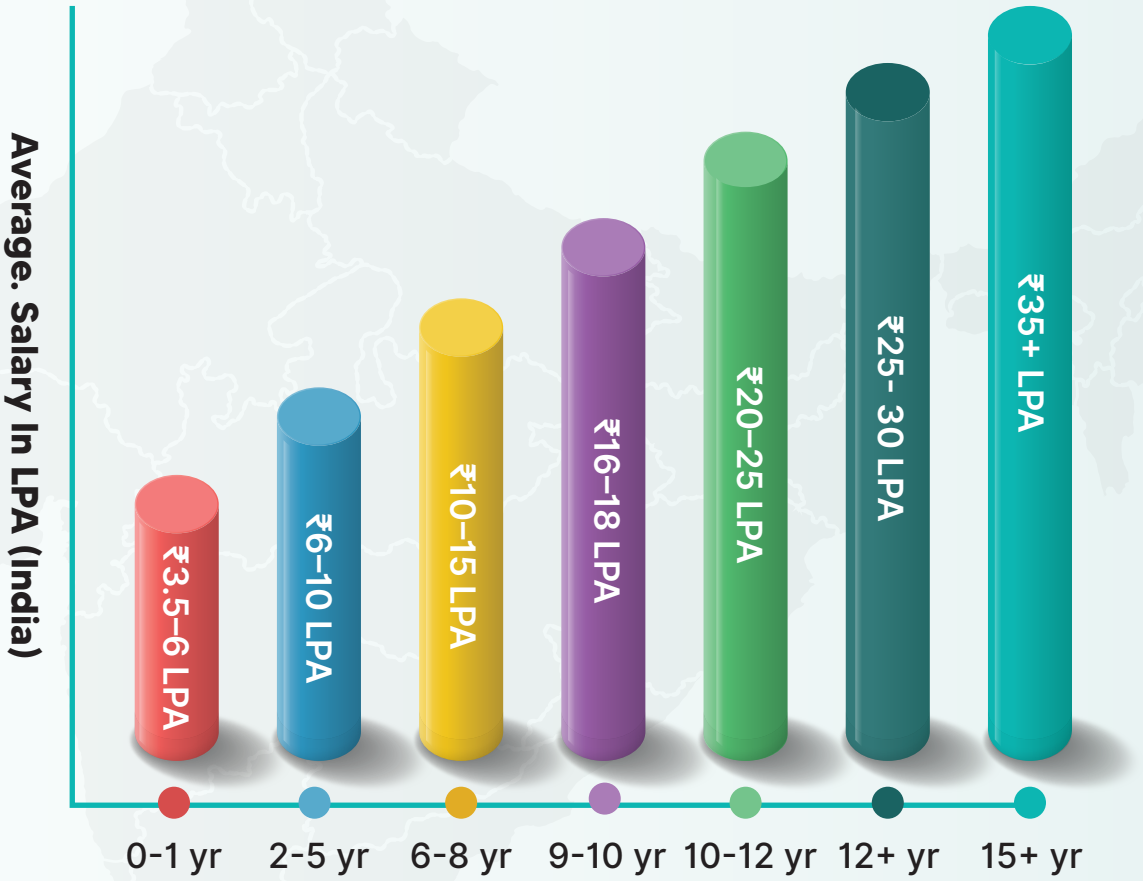


WHO SHOULD JOIN THIS PROGRAM?

- Engineering students (ECE, EEE, CSE, IT) who want hands-on experience with hardware, sensors, and microcontrollers.
- Beginners aspiring to build a career in embedded systems or IoT with no prior experience required.
- Working professionals in electronics or software looking to upskill into the fast-growing IoT industry.
- Diploma students in electronics/electrical seeking practical, industry-ready skills.
- Tech enthusiasts and hobbyists who want to build real-world IoT devices and smart automation projects.
- Embedded firmware aspirants wanting to strengthen programming skills in C/C++ and microcontroller concepts.
- IoT product builders and innovators planning to create smart home, automation, or industrial IoT solutions.
- Anyone preparing for embedded/IoT job roles such as Embedded Engineer, IoT Developer, Hardware Engineer, or R&D Engineer.



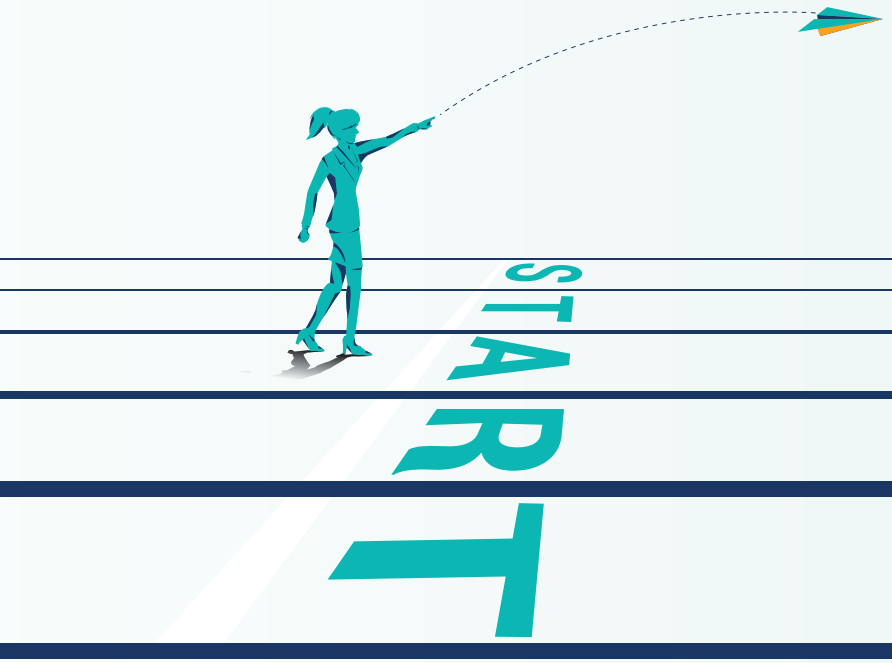
CAREER ROLES & SALARIES



- Embedded Systems Engineer
- IoT Embedded Developer
- IoT Application Developer
- Embedded Linux Engineer
- Technical Lead (Embedded/IoT)
- Embedded Software Engineer
- PCB Design Engineer
- Systems Engineer
- R&D Engineer (Embedded/IoT)
- IoT Engineer

CAREER TRAJECTORY

Experience Level	Average Salary (India)
(0–2 yr)	₹3.5–6 LPA
(2–4 yr)	₹6–10 LPA
(5–7 yr)	₹10–15 LPA
(8+)	₹16-18 LPA
(12+ yr)	₹25–30 LPA



START CAREER WITH EMBEDDED & IoT COURSE

PLACEMENT SUPPORT

WE

400+ Hiring Partner Opportunities

Access a vast network of 440+ hiring partners, giving you more chances to land interviews and secure your dream job.

Company-wise Interview Questions

Practice with real interview questions from leading companies to boost confidence and improve your selection chances.

Resume, LinkedIn & Portfolio Building

Get expert guidance to create a powerful resume, optimize LinkedIn, and build a portfolio that highlights your skills to employers.

Alumni Referral Network

Leverage our alumni working in top companies to get trusted referrals and unlock hidden job opportunities.



WANT YOU

SUPPORT & CAREER GUIDANCE

Discover Strengths & Interests

Every learner has unique talents, skills, and passions but many are unsure how to identify and apply them in their career. That's where we step in. Through assessments, mentoring sessions, and hands-on exercises, we help learners discover what they're truly good at and where their genuine interests lie. This self-awareness becomes the foundation for making smarter career choices, ensuring that learners not only land a job but also enjoy long-term growth and satisfaction in their chosen field.

Explore Career Paths

The world of work is full of opportunities, but without proper guidance, learners often feel lost. We provide detailed insights into diverse industries, in-demand roles, and future career trends, helping learners explore multiple options before deciding on the right path. By understanding the scope, responsibilities, and growth potential of each career track, learners can confidently pursue the path that matches both their skills and aspirations, leading to meaningful and sustainable career success.

Prepare for Interviews

A strong resume can get you an interview, but it's the preparation that gets you the job. Our structured interview training includes mock sessions, role-specific practice questions, and personalized feedback to polish communication and problem-solving skills. Learners gain confidence in tackling tough questions, handling pressure, and presenting themselves professionally. With this kind of preparation, they enter interviews not just as applicants, but as top contenders ready to impress employers.

Build Professional Portfolios

In today's competitive job market, employers want proof of skills, not just promises. We guide learners in building professional portfolios that showcase real projects, case studies, and achievements. These portfolios act as a visual proof of their abilities, giving them a strong edge over other candidates. Whether it's tech projects, design work, research, or case studies, a well-crafted portfolio makes a lasting impression on recruiters and ensures learners are remembered long after the interview ends.

How to Enroll & Next Steps

ENROLLMENT PROCESS

Step 1: Free Consultation

- Book a free career counseling session
- Understand your career goals and fit



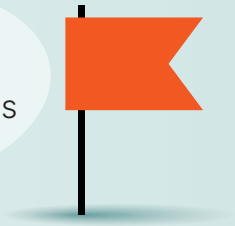


Step 2: Trial Class

- Attend one free trial class
- Experience our teaching methodology

Step 3: Enrollment

- Choose your batch timing
- Complete admission formalities





Step 4: Begin Learning

- Start your transformation journey
- Access lifetime support system

Next Batch Details:
Batch Start Date: Every Monday

Class Timings:
Weekday Batch: **7:00 PM - 9:00 PM**
Weekend Batch: **10:00 AM - 1:00 PM**

Mode:
Live Online + Offline (Noida Center)

Batch Size:
Limited to 15 students

PROGRAM FEE

₹31,999 (Including GST)

Scholarship Available Upto 30%

(Scholarship is granted based on the candidate's Work Experience, Statement of Purpose,and Academic Background.)

Payment Options

01 Full Fee Payment

1. Total Fee Payment: ₹31,999 (Including GST)
2. Admission Fees (payable on enrolment): ₹1,000

02 Installment Plan

1. Total Fee Payment: ₹31,999 (Including GST)
2. Admission Fees (payable on enrolment): ₹1,000
3. No. of Installments: 3 & 4 Months

Note:

- *Full fee will be divided into 3 or 4 monthly installments after deducting the scholarship awarded*
- *Contact your counselor for more details*

03 Loan & EMI Option

1. Total Fee Payment: ₹31,999 (Including GST)
2. Admission Fees (payable on enrolment): ₹1,000
3. Loan Amount: ₹30,999
4. No-Cost EMI Available: 3 Months & 4 Months

Note:

- *Full fee will be divided into 3 or 4 monthly EMIs after deducting the scholarship awarded*
- *Choose from our financing partners for interest-free EMI options*
- *Contact your counselor for more details*

Embedded & IoT Certification

Why These Certifications Matter?

- Industry-Recognized Credential endorsed by NASSCOM's FutureSkills Prime platform.
- Validates job-ready skills aligned with India's National Skills Framework (NSQF).
- Enhances employability with government-backed digital skills certification.
- Mapped to industry standards set by top tech companies and IT bodies.
- Boosts professional credibility with a nationally accepted qualification.
- Certifies mastery of emerging technologies demanded by leading employers.
- Supports career growth with proof of competency in specialized domains.
- Eligible for government incentives under the FutureSkills Prime program (where applicable).
- Adds weight to resumes & LinkedIn profiles for stronger job visibility.
- Recognized across IT, analytics, and AI-driven industries for recruitment and upskilling.



THE IoT ACADEMY VS OTHERS

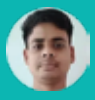
Factors	The IoT Academy	Other Institutes
Founder Background	IIT Alumni	Varied
Batch Size	10-15 students	50-100 students
Fee Structure	₹31,999 (including GST)	₹50,000-₹80,000
Tools Training	Comprehensive	Limited/None
1:1 Mentorship	Daily sessions	Group sessions only
Salary Hike	78% Average	30-50% average
Industry Partnerships	IIT Collaboration	Limited
Live Projects	Real client work	Simulated projects

Why Students
CHOOSE US OVER COMPETITORS



Hemant Raj
Placed in MORPHEDO

The IoT Academy's Embedded System course was a game-changer for me! The hands-on projects helped me understand complex concepts easily. I feel confident in my skills now and ready for a career in IoT!



Dharmendra
Placed in AEONX DIGITAL

I loved the Embedded System certification course! The instructors were knowledgeable and supportive. The practical experience I gained was invaluable. I highly recommend this course to anyone looking to enter the IoT field!



Ritik Verma
Placed in MORPHEDO

Completing the Embedded System course at The IoT Academy was an amazing experience. The curriculum was well-structured, and I learned so much. I now have the skills to pursue my dream job in technology!



READY TO TRANSFORM YOUR CAREER?

DON'T WAIT - THE DIGITAL FUTURE IS NOW!

Join thousands of successful professionals who transformed their careers with The IoT Academy. ***Limited seats available.***

ENQUIRE FOR REGISTRATION TODAY!

The IoT Academy - Where Technology Meets Opportunity
Empowering India's Digital Workforce Since 2017



THANK YOU

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